

Assignment 1: Neural Networks

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1. Executive Summary

The goal of this assignment was to improve and evaluate a neural-network model that classifies IMDB movie reviews as positive or negative. I compared changes in network depth, number of hidden units, loss function, activation function, and dropout regularization. The main finding was that increasing model complexity did not improve validation performance. A relatively small model with two 8-unit hidden layers, tanh activation, and 20% dropout produced the highest mean validation accuracy and lowest variability among the three leading candidates tested. After model selection, this model achieved 88.05% accuracy on the untouched test set.

2. Data and Modeling Procedure

The TensorFlow/Keras IMDB dataset contains 25,000 training reviews and 25,000 test reviews. The vocabulary was limited to the 10,000 most frequent words. Each review was represented as a 10,000-element binary vector indicating whether each word was present. During model development, 15,000 reviews were used to train the models and 10,000 were reserved for validation. The separate 25,000-review test set was not used for model selection.

The baseline model contained two hidden layers with 16 units per layer. ReLU, an activation function that determines how information is transformed within the network, was used in the hidden layers, followed by a sigmoid output that produced the positive-versus-negative classification. RMSprop was used to update the model during training, while binary crossentropy measured how well the model distinguished positive from negative reviews. A batch size of 512 was used. Models were initially trained for 20 epochs so that both performance and overfitting could be observed.

I used a controlled comparison: when one model setting was evaluated, the other major settings were kept unchanged. I tested one, two, and three hidden layers; 8, 16, 32, and 64 hidden units; binary crossentropy versus mean squared error (MSE); ReLU versus tanh activation; and dropout rates of 0.2 and 0.5. Tanh is an alternative activation function, while dropout is a regularization method that temporarily removes a fraction of network connections during training to help reduce overfitting. Validation accuracy was the main performance measure used for model comparison.

3. Baseline Performance and Overfitting

The baseline model reached its best validation accuracy of 88.80% at epoch 4 and its lowest validation loss of 0.2790 at the same epoch. After the early epochs, training accuracy continued to increase while validation accuracy stopped improving and validation loss increased. This is evidence of overfitting: the model continued to fit the training data more closely without improving its performance on unseen validation reviews.

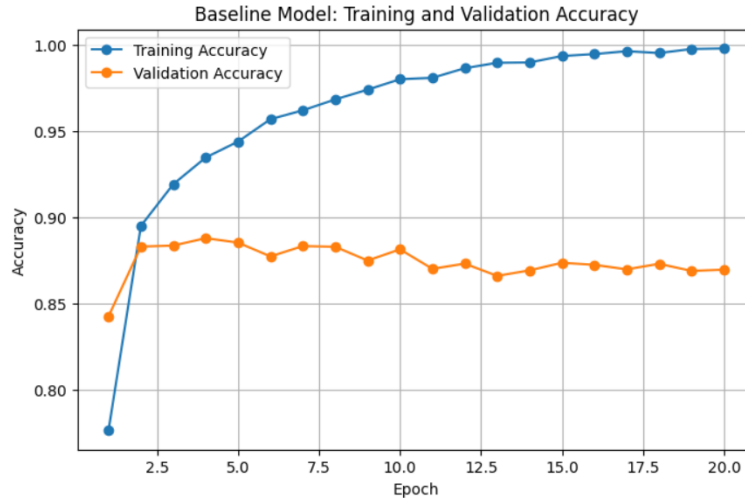


Figure 1. Training and validation accuracy for the baseline model.

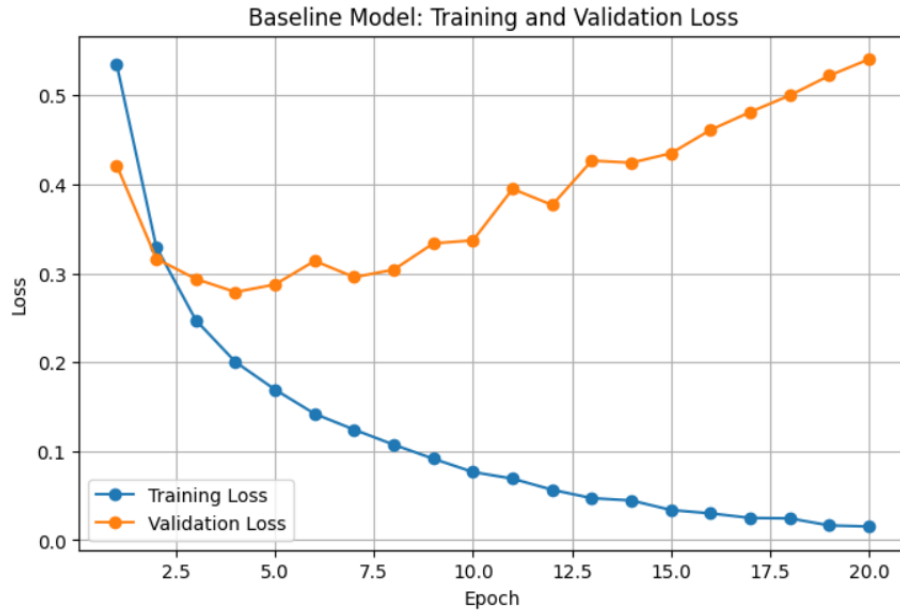


Figure 2. Training and validation loss for the baseline model.

4. Experimental Results

Table 1 summarizes the model comparisons. The values shown are the best validation accuracies observed during each 20-epoch training run.

Model	Layers	Units	Activation	Loss	Dropout	Val. Acc. (%)	Epoch
Baseline	2	16-16	ReLU	Binary CE	0	88.80	4
1 Hidden Layer	1	16	ReLU	Binary CE	0	88.83	3
3 Hidden Layers	3	16-16-16	ReLU	Binary CE	0	88.64	5
8 Units	2	8-8	ReLU	Binary CE	0	88.96	5
32 Units	2	32-32	ReLU	Binary CE	0	88.26	5
64 Units	2	64-64	ReLU	Binary CE	0	88.59	4
MSE Loss	2	16-16	ReLU	MSE	0	88.66	6
Tanh Activation	2	16-16	Tanh	Binary CE	0	88.99	3
Dropout 0.2	2	16-16	ReLU	Binary CE	0.2	88.92	4
Dropout 0.5	2	16-16	ReLU	Binary CE	0.5	88.71	6
Combined Model	2	8-8	Tanh	Binary CE	0.2	88.96	5

Table 1. Summary of the hyperparameter experiments.

Network depth had little effect on validation accuracy. One hidden layer reached 88.83%, the two-layer baseline reached 88.80%, and three hidden layers reached 88.64%. Therefore, adding depth did not improve performance in these tests. Network width showed a similar pattern: the smaller 8-8 model achieved 88.96%, while the 32-32 and 64-64 models achieved 88.26% and 88.59%. More parameters did not translate into better generalization.

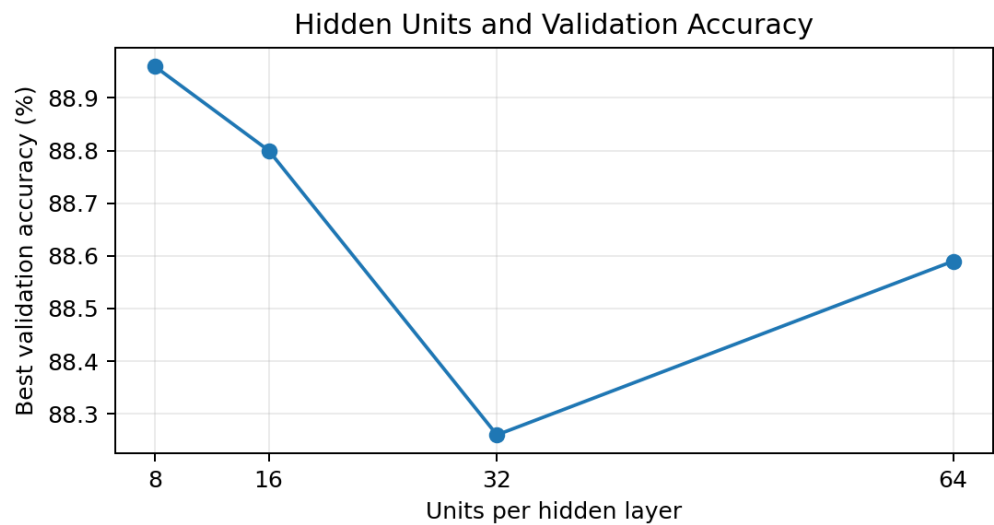


Figure 3. Effect of hidden-layer width on validation accuracy.

Changing the loss function to MSE resulted in 88.66% validation accuracy, slightly below the 88.80% binary-crossentropy baseline. Using tanh instead of ReLU produced the highest single-run validation accuracy, 88.99%. Moderate dropout of 0.2 achieved 88.92%, while stronger dropout of 0.5 achieved 88.71%. These results suggest that moderate regularization was more suitable than the stronger dropout setting for this dataset.

5. Model Selection and Stability

The strongest single-run results differed by only a few hundredths of a percentage point. Because neural-network training depends partly on random weight initialization and dropout, selecting a final model from one run could be misleading. I therefore repeated three leading configurations using three fixed random seeds and compared their mean validation accuracy and variability.

Candidate Model	Mean Val. Acc. (%)	SD (percentage points)	Mean Best Epoch
ReLU 8-8	88.810	0.125	6.333
Tanh 16-16	88.780	0.122	3.000
Tanh 8-8 + Dropout 0.2	88.967	0.071	4.667

Table 2. Stability comparison of the three leading candidate models.

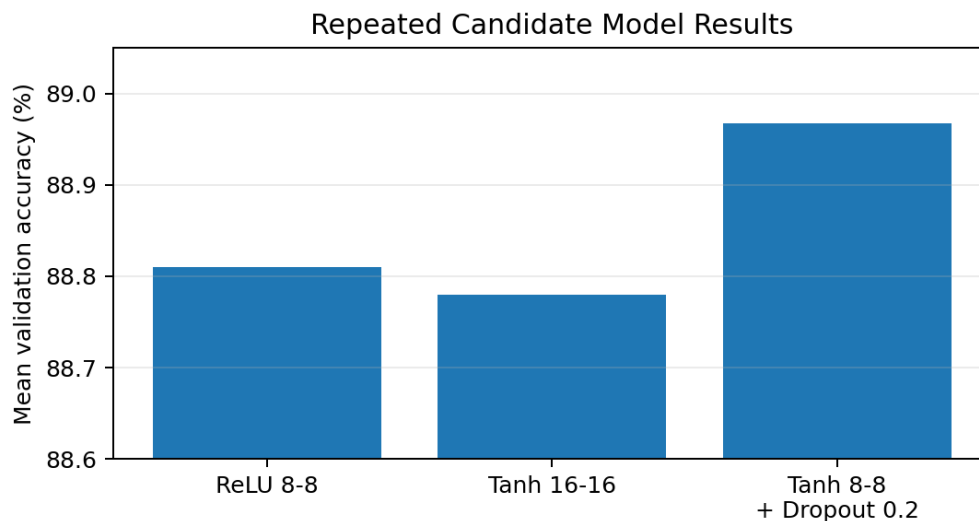


Figure 4. Mean validation accuracy of the repeated candidate models.

The combined 8-8 tanh model with 0.2 dropout had the highest mean validation accuracy (88.967%) and the lowest standard deviation (0.071 percentage points). Although the tanh 16-16 model produced the highest individual validation run earlier, the repeated comparison favored the combined model in both average validation accuracy and stability. However, the differences among the leading models were small, and the comparison was based on only three random seeds, so the ranking should be interpreted as suggestive rather than definitive. Based on these results, the combined model was selected for final evaluation.

6. Final Performance and Recommendation

After model selection was complete, the selected architecture was retrained using all 25,000 training reviews. Five epochs were used because the repeated validation runs identified a mean best epoch of approximately 4.67. The final model was then evaluated once on the untouched 25,000-review test set.

Final Metric	Result
Test accuracy	88.05%
Test loss	0.3005
Selected architecture	8-8, tanh, dropout 0.2
Optimizer / loss	RMSprop / binary crossentropy

Table 3. Final model and test-set performance.

Recommendation: for this IMDB classification task, the results support using a smaller regularized network rather than simply increasing network depth or width. The experiments did not show a performance benefit from the larger 32-32 or 64-64 networks or from adding a third hidden layer. The selected 8-8 tanh model with moderate dropout provided the strongest average validation performance among the repeated candidates while using a relatively simple architecture.

7. Conclusions

This assignment showed that model complexity alone does not guarantee better prediction. The baseline model began to overfit after the early epochs, and larger or deeper networks did not improve validation accuracy. Tanh activation and moderate dropout were useful changes, while MSE and stronger dropout did not improve performance relative to the baseline. Repeating the leading candidates provided a more reliable basis for model selection than relying on one training run. The final selected model achieved 88.05% accuracy on unseen test reviews, which was reasonably close to its validation performance.

The complete Python code, training histories, and raw outputs are provided separately in the submitted Jupyter notebook (.ipynb), rather than in this summary report.